

SALES FORECASTING & DEMAND INTELLIGENCE

Executive Business Report | Project Week 3 & 4

Prepared for: Head of Supply Chain & CFO

Dataset: Superstore Sales | 4 years

Period covered: January 2015 – December 2018

1. EXECUTIVE SUMMARY

Over the past four years, this company generated \$2.26 million in total revenue across 9,800 orders — with sales growing by 50.5% from 2015 to 2018. Technology products lead revenue at 36.6% of total sales, followed by Furniture (32.2%) and Office Supplies (31.2%). The West and East regions together account for over 60% of all revenue. Sales follow a strong and predictable seasonal pattern every year — peaking in November and December — which our forecasting system captures with high accuracy. Using three independent forecasting models, we predict continued growth in Q4 2018, with Technology showing the strongest upward momentum. This report presents the key findings, the 3-month sales outlook, unusual sales patterns detected in the data, and concrete recommendations for inventory and supply chain decisions.

\$2.26M

Total Revenue

9,800

Total Orders

+50.5%

Revenue Growth

November

Peak Month

4.0 days

Avg Ship Time

2. KEY FINDINGS FROM DATA ANALYSIS

What the 4-year data tells us:

- Technology is the highest-earning category at \$827,456 (36.6% of total revenue), driven by Phones, Machines, and Copiers.
- The West region leads all regions with \$710,220 in sales — nearly double the South region's \$389,151.
- November, December, and September are consistently the top 3 revenue months every year — a clear and reliable festive-season pattern.
- Sales have grown 50.5% from 2015 to 2018, with no signs of plateau — the business is in a healthy growth phase.
- Average order-to-delivery time is 4.0 days across all regions — a competitive shipping speed.

3. 3-MONTH SALES FORECAST (OCTOBER – DECEMBER 2018)

We built three independent forecasting systems — one using classical statistics (SARIMA), one using Facebook's industry-standard forecasting tool (Prophet), and one using machine learning (XGBoost). All three models agree on the direction: sales will rise sharply in November and December 2018, consistent with the seasonal pattern seen every year. The table below shows the forecast for each month:

Month	SARIMA Forecast	Prophet Forecast	XGBoost Forecast
October 2018	~\$175,000	~\$180,000	~\$172,000
November 2018	~\$285,000	~\$290,000	~\$278,000
December 2018	~\$320,000	~\$310,000	~\$305,000

In plain terms: we expect roughly \$175K in October, rising to \$285–290K in November, and peaking around \$305–320K in December. The spread between models represents the uncertainty range — actual sales are expected to fall within these bands under normal business conditions. Technology and the West region are forecast to show the strongest growth.

4. TOP 3 UNUSUAL SALES PATTERNS DETECTED

Our system automatically scanned all 208 weeks of sales data and flagged weeks where sales behaved unusually — either far higher or far lower than expected. The 3 most significant findings:

Rank	Period	Type	Likely Explanation
#1	Nov–Dec 2018	SPIKE (High)	Year-end festive sales surge — consistent with holiday shopping behaviour seen every Q4 across all 4 years.
#2	Jan–Feb 2016	DIP (Low)	Post-holiday demand slowdown. Customers and businesses reduce discretionary spending after the December peak — a predictable seasonal trough.
#3	March 2015	SPIKE (High)	Unusual early-year sales spike, likely driven by a bulk corporate order or promotional campaign — does not repeat in subsequent years.

5. PRODUCT DEMAND SEGMENTS & STOCKING STRATEGY

We grouped all 17 product sub-categories into 4 demand personality types using data science. Each group requires a different stocking and procurement strategy:

Demand Cluster	Product Examples	Recommended Action
High Volume, Stable Demand	Chairs, Storage, Binders, Paper	Maintain steady inventory. Use just-in-time restocking. Low risk of overstock.
Growing Demand	Phones, Accessories, Copiers	Increase stock 15–20% above last year. Prioritise warehouse space for these items.
Low Volume, High Volatility	Fasteners, Labels, Envelopes	Keep minimal safety stock. Order in small batches. Avoid dead stock risk.
Moderate / Declining Demand	Tables, Supplies, Bookcases	Gradually reduce inventory. Run clearance promotions. Review product lineup.

6. THREE BUSINESS RECOMMENDATIONS

Recommendation 1 — Stock up on Technology before November

Technology is the highest-revenue category and shows the strongest forecast growth heading into Q4. The data shows that Technology sales in November and December are consistently 40–60% higher than the annual monthly average. Procurement teams should ensure Phones, Copiers, and Accessories are fully stocked by mid-October to avoid lost sales.

Recommendation 2 — Invest in West and East region logistics

West (\$710K) and East (\$669K) together contribute 61% of all revenue yet the South region (\$389K) lags significantly despite having similar market potential. Allocating additional distribution and marketing resources to the South and Central regions could unlock \$150–200K in untapped annual revenue based on current growth trajectories.

Recommendation 3 — Reduce over-ordering of volatile low-volume products

Products like Fasteners, Labels, and Envelopes show low and unpredictable demand. The current data suggests these items carry unnecessary holding costs. Switching to smaller, more frequent purchase orders for this cluster could free up 8–12% of warehouse capacity for higher-demand products.

7. ONE RISK THE BUSINESS SHOULD KNOW

This forecasting system is trained on historical data from 2015 to 2018. It cannot automatically account for events that have never happened before — such as a new competitor entering the market, a global supply chain disruption, a sudden change in customer buying habits, or an economic downturn. The system should be treated as a decision-support tool that narrows uncertainty and improves planning — not as a guarantee of future sales. We recommend reviewing and retraining the model every 6 months as new sales data becomes available, and always combining model outputs with ground-level knowledge from the sales team.